

MiniMax-M3 × Kimi K3 × Opus 4.8

● REFRESH · K3-AUTHORED

MISSION M3-VS-KIMI-K3-2026-07-17 · REVISION 2 · STELLA
INTELLIGENCE + COMPTROLLER + OSINT CELL

Part I is the refresh — researched and written by Kimi K3, the model now running this harness. Part II is the original record, written by MiniMax-M3 eleven hours earlier. The comparison is demonstrated, not just described.

LOOP R2 · TERMINAL STATE CONVERGED (7/8 GAPS CLOSED, 1 HONEST NULL) · DEDUPE
SHA256("M3-VS-KIMI-K3"+"2026-07-17"+R2)[:16] · IDEMPOTENT OVERWRITE-ON-SAME-
DEDUPE-KEY · AAR ~/ .CLAUDE/CACHE/CORPORATE/AARS/M3-VS-KIMI-K3-R2-2026-07-17.MD

PART I — THE REFRESH K3

§R0 EXECUTIVE VERDICT (R2)

v1 verdict (M3, 11 hours earlier): “Don’t wholesale switch. Pilot K3 on long-horizon synthesis + code gen; keep Opus 4.8 for agentic production.”
Owner superseded: **switched the harness default to Kimi K3 via OpenRouter the same day**, with the eval harness as the measurement layer.

r2 verdict (K3, now): The switch is **defensible on capability grounds** — the release-day claim “zero agentic benchmarks” is already stale (Terminal-Bench 2.1 **88.3**, above Opus 4.8’s 84.6 — vendor-published, unreplicated). The cost story is **more nuanced than the hype in both directions**: K3 is ~40–48% cheaper than Opus 4.8 per task, but **MiniMax-M3’s API is ~10× cheaper than K3** (\$0.30/\$1.20 vs \$3/\$15), and K3’s always-on max reasoning produces verbose bills (a trivial prompt cost \$0.25 in Willison’s test). For STELLA’s harness both M3 and K3 arrive **subscription-bundled** — so the prices below matter for the OpenRouter eval runs and future API budgeting, not for today’s marginal cost, which is \$0 either way.

Posture unchanged in shape, updated in content: `engage-with-batna` — the engagement happened; the BATNA is a warm rollback path to M3 plus the eval-harness gate before Phase 2.

What r2 changes: 7 of v1’s 8 open gaps are now **closed** (M3 pricing, K3 context, K3 agentic benchmarks, production reports, open-weights license, Fable 5 pricing, 24h news sweep). One **honest null** remains (K3 active params — officially undisclosed). One v1 figure is **corrected** (K3 “SWE-bench 76.8%” was never official — do not use).

§R1 WHAT CHANGED SINCE V1 (THE SAME DAY)

#	EVENT	CONSEQUENCE
1	Owner set harness default to Kimi K3 (moonshotai/kimi-k3 [1m])	The comparison stopped being hypothetical. This refresh is itself a K3 agentic run on the STELLA stack.
2	M3 pricing surfaced (official MiniMax docs)	v1's hardest gap closed. Reframes §R4 cost analysis entirely.
3	Moonshot published agentic benchmark table	v1's #1 risk ("zero agentic benchmarks") downgraded from <i>unknown</i> to <i>vendor-claimed, unreplicated</i> .
4	Table 5 priced publicly (\$10/\$50)	The top-tier anchor now has a meter.
5	First-24h hands-on reports landed (Willison, r/LocalLLaMA, HN ~572 pts)	Real production texture: ~30 t/s, verbose reasoning, "one of the best I've ever seen."

§R2 GAP-CLOSURE REPORT (INTELLIGENCE_OFFICER)

GAP (FROM V1 §9)	STATUS	FINDING	CONF.
1. MiniMax-M3 public pricing/specs	CLOSED	\$0.30 in / \$1.20 out per MTok ($\leq 512K$ ctx); \$0.60/\$2.40 ($> 512K$); cache-read \$0.06/\$0.12; priority tier 1.5×. Context 1M tokens , max output 512K (default 128K), model ID <code>minimax-m3</code> , text+image+video input.	High
2. K3 context window	CLOSED	1M tokens (1,048,576) ; max completion default 131,072, configurable to 1M. Fixed sampling temp=1.0, top_p=0.95. Flat pricing, no ctx tiering.	High
3. K3 active params	STILL-OPEN (honest null)	Officially undisclosed . Confirmed: 16 of 896 experts active (Stable LatentMoE). Third-party estimate ~80–90B active (extrapolated, speculative). The circulating “~50B” is naive expert-only math — do not use .	High that it's undisclosed
4. K3 agentic tool-calling benchmarks	CLOSED	Official table (max reasoning): Terminal-Bench 2.1: 88.3 (Fable 5: 84.6, GPT-5.6 Sol: 88.8, Opus 4.8: 84.6) · MCP Atlas: 84.2 · BrowseComp: 91.2 · DeepSWE 67.5 · FrontierSWE 81.2 · SWE Marathon 42.0 · HLE-Full w/tools 56.0. SWE-bench Verified: not published .	High (official); Med (unreplicated)
5. K3 production reports	CLOSED	Willison: 95-token prompt → 16,658 output tokens (13,241 reasoning) = \$0.25 ; ~85-token hidden system prompt; vision alt-text “very good”; agentic/long-conv untested. Moonshot's own caveats: dropping historical <code>reasoning_content</code> → generation “highly unstable”; “excessive proactiveness” on long-horizon tasks; “noticeable gap in user experience” vs Fable 5 / GPT-5.6 Sol. r/LocalLLaMA: ~30 t/s , one frontend task took 35 min, quality “one of the best I've ever seen.”	High

GAP (FROM V1 \$9)	STATUS	FINDING	CONF.
6. K3 open weights	CLOSED	On track 2026-07-27 . License Modified MIT (commercial OK). No HF repo yet as of 07-17. Technical report ships with weights. Deployment guidance: 64+ accelerator supernodes; MXFP4 weights / MXFP8 activations.	High
7. Fable 5 pricing	CLOSED	\$10 / \$50 per MTok ; cache write \$12.50 (5-min) / \$20 (1-hr), hits \$1; US-only inference 1.1x. New tokenizer ~30% more tokens vs older Claudes (real cost multiplier). <code>claude-fable-5</code> on Claude Platform/AWS/GCP/Foundry.	High
8. New-in-24h	CLOSED	See \$R5 + \$R7. No claim walk-backs found; M3 pricing unchanged; no Opus 4.8 changes.	High

Convergence: 7/8 closed, 1 honest null. Sources: platform.minimax.io, kimi.com/blog/kimi-k3, platform.kimi.ai docs, simonwillison.net, anthropic.com/claude/fable, platform.claude.com pricing, artificialanalysis.ai, tosea.ai, aireleasetracker.com, the-decoder, r/LocalLLaMA, HN.

\$R3 THE M3 REPRICING — V1'S BIGGEST GAP WAS HIDING THE CHEAPEST MODEL

v1 treated M3 as a black box and framed the decision as “K3 (pilot) vs Opus 4.8 (production).” The surfaced pricing rearranges the board:

MODEL	INPUT \$/M	OUTPUT \$/M	CACHE-READ \$/M	CONTEXT	NOTES
MiniMax-M3	\$0.30	\$1.20	\$0.06	1M	Cheapest metered model in this dossier by an order of magnitude
Kimi K3	\$3	\$15	\$0.30 (mechanics undocumented)	1M	“Sonnet-priced, not cheap-Chinese-priced” — most expensive Chinese-lab model to date (the-decoder)
Opus 4.8 std	\$5	\$25	\$0.50	200K	1,024-token cache minimum; mid-conv system messages
Opus 4.8 fast	\$10	\$50	—	200K	Buys latency, not savings
Fable 5	\$10	\$50	\$1	—	New tokenizer $\approx$ +30% tokens $\rightarrow$ real cost multiplier

Read it correctly: K3’s “cheap” framing only holds against Opus and Fable. Against M3, K3 is **10–12.5× more expensive per token**. The justification for K3 over M3 is therefore *never* price — it is capability tier (Terminal-Bench 88.3, AA Intelligence Index 57 vs 56 over Opus 4.8, GPQA-Diamond 93.5 — all vendor/arena-sourced, unreplicated; see §R5). Meanwhile M3’s \$0.30/\$1.20 makes it the rational metered choice for **bulk, lower-stakes workloads** if STELLA ever pays per-token.

The subscription caveat that governs today: Christian consumed M3 — and now consumes K3 — as a **bundled harness entitlement**, marginal cost \$0. The meters above govern (a) the OpenRouter eval-harness runs, (b) any direct-API workloads, (c) the vendor-economics question of what the subscription is actually worth.

\$R4 COST ANALYSIS (COMPTROLLER) — RECONCILED
 WITH M3 PRICING

Price inputs: K3 \$3/\$15 (**VERIFIED**) · Opus 4.8 \$5/\$25 std, \$10/\$50 fast (**VERIFIED**) · Opus cache read \$0.50/M at 90% discount, 1,024-token minimum (**VERIFIED**) · K3 cache read \$0.30/M, mechanics undocumented (**price VERIFIED / mechanics UNKNOWN**) · M3 \$0.30/\$1.20, cache \$0.06 (**VERIFIED r2**). All token volumes **ESTIMATE**.

PER-WORKLOAD COST (PER RUN)

WORKLOAD	TOK IN/OUT (EST.)	M3	K3	OPUS STD	OPUS FAST
OJP application kit	3K / 1.5K	\$0.003	\$0.032	\$0.053	\$0.105
AAR generation	4K / 1K	\$0.002	\$0.027	\$0.045	\$0.090
OSINT dossier synthesis	30K / 8K	\$0.019	\$0.210	\$0.350	\$0.700
Strategic posture analysis	6K / 2.5K	\$0.005	\$0.056	\$0.093	\$0.185
Workbook codegen	15K / 6K	\$0.012	\$0.135	\$0.225	\$0.450
STELLA 53-agent dispatch (full sweep)	120K / 25K	\$0.066	\$0.735	\$1.225	\$2.450
PDF pipeline (content gen only)	5K / 3K	\$0.005	\$0.060	\$0.100	\$0.200
Long-horizon research synthesis	80K / 15K	\$0.042	\$0.465	\$0.775	\$1.550
Recruiter counter-strategy	4K / 1.5K	\$0.003	\$0.035	\$0.058	\$0.115

(M3 column computed at \$0.30/\$1.20 ≤512K tier — all listed workloads fit. Comptroller’s note: K3 lands ~40% cheaper than Opus-std on these shapes, not the headline 48% — Willison’s figure reflects his benchmark’s token mix; K3’s edge widens as output share grows.)

102-DAY SPRINT TCO (ESTIMATE: 50M IN / 10M OUT PER MONTH;
3.4 MONTHS)

SCENARIO	MONTHLY	102-DAY TOTAL
M3-only (metered)	\$27	\$92
K3-only	\$300	\$1,020
Opus-std-only (no cache)	\$500	\$1,700
Opus-std with 50% input cache-hit	\$387.50	\$1,318
Mixed: 60% vol K3 + 40% Opus-std cached	\$335	\$1,139

Mixed routing lands within 12% of K3-only while keeping agentic workloads on proven infrastructure. M3-only is 11× cheaper than K3-only — at a capability tier the benchmarks say is lower; the gap is the price of the frontier.

CACHE ECONOMICS — WHERE OPUS CLAWS IT BACK

Opus 4.8’s 1,024-token cache minimum (**VERIFIED**) is decisive for STELLA: officer system prompts + standing orders (~2–8K tokens) qualify. A 53-agent sweep repeating a 6K prefix: uncached input = $318K \times \$5/M = \1.59 ; cached = one write ($6K \times \$6.25/M$ write price $\approx \$0.04$) + 52 reads $\approx \$0.19$ — an ~88% input saving that flips the **input-side** ranking in Opus’s favor despite K3’s lower sticker. Honest total: cached-Opus sweep $\approx \$0.815$ ($\$0.19$ in + $\$0.625$ out) still loses to K3’s $\$0.735$ on this workload shape — which is exactly why the routing rule below is input-scoped, not sweep-scoped. **Break-even derivation:** Opus effective input = $\$5(1-f) + \$0.50f = \$3$ (K3’s price) at $f = 0.44$. Above ~45% cache-hit fraction, cached Opus input undercuts K3’s base input — and STELLA dispatches run 70–90% repeated prefix. K3 lists a $\$0.30/M$ cached-input price but Moonshot publishes no minimum, TTL, or invalidation behavior, and K3 lacks mid-conversation system messages — **assume Opus cache = reliable, K3 cache = bonus if it hits.**

ROUTING RULE (PURELY \$, PER COMPTROLLER)

- **Output-heavy long synthesis (>5K output, one-shot):** route **K3**. Cheaper output + 21% fewer output tokens compounds; cache irrelevant on one-shots.
- **Agent loops with ≥45% repeated input (static prefix ≥1,024 tok):** route **Opus-std** if metered — cached input beats K3’s base price.
- **Bulk low-stakes workloads, if ever metered:** route **M3-API** — $\$0.30/\1.20 is the floor of this market.
- **Small routine tasks (<\$0.10/run either way):** route on quality/latency, not dollars — the delta is noise.
- **Governing caveat:** inside the harness subscription all of the above is \$0-marginal; the rule activates the day STELLA pays per token.

\$R5 K3’S AGENTIC PICTURE — WHAT EXISTS NOW
THAT DIDN’T AT V1

v1’s #1 stated risk was “**zero published agentic tool-calling benchmarks.**” That was true at ~19:00 and stale by evening. Moonshot’s release table (all at max reasoning):

BENCHMARK	K3	FIELD REFERENCE	READ
Terminal-Bench 2.1	88.3	GPT-5.6 Sol 88.8 · Fable 5 84.6 · Opus 4.8 84.6	K3 is genuinely in the frontier cluster on terminal-agentic work
MCP Atlas	84.2	—	Directly relevant to STELLA's MCP-heavy stack
BrowseComp	91.2	—	Web-research agentic (intelligence_officer's home turf)
DeepSWE	67.5	—	Vendor's alternative to SWE-bench
FrontierSWE	81.2	—	Vendor's alternative to SWE-bench
SWE Marathon	42.0	—	Long-horizon SWE
HLE-Full w/ tools	56.0	—	Humanity's Last Exam, tool-augmented
SWE-bench Verified	not published	Opus 4.5: 80.9	v1's "K3 76.8%" was never official — corrected, do not use

New-in-24h capability items (intelligence_officer): Artificial Analysis Intelligence Index v4.1: **K3 = 57 (#3–4/189) vs Opus 4.8 = 56**; Fable 5 = 60, GPT-5.6 Sol(max) = 59. K3 burned 130M tokens in that eval (2× avg verbosity) at ~\$0.94/index task. Frontend Code Arena: **K3 #1 at 1,679** over Fable 5 (1,631) — a 17-place jump from K2.6, with a documented voter-bias caveat toward flashy 3D/Three.js. GPQA-Diamond **93.5** — beats Fable 5 (92.6), trails GPT-5.6 Sol (94.1). HN front page ~572 pts / 300+ comments; sentiment “China is 6 days behind, not 6 months.”

Honest caveats: every number in this section is **vendor-published or arena-sourced; none independently replicated**. Moonshot's *own* release caveats concede “excessive proactiveness” on long-horizon tasks and a “noticeable gap in user experience” vs Fable 5 / GPT-5.6 Sol. The eval harness (`run_eval.py` + `score.py`) exists precisely to convert vendor claims into STELLA-workload evidence.

§R6 DAY-ONE-ON-K3 FIELD NOTES

Bounded honesty: this is one working session, not a production track record. What can be said on day one:

- This refresh is itself the first data point.** K3 dispatched two STELLA officers (intelligence + comptroller) in parallel, ingested their structured returns, and is synthesizing a gated, NDA-scrubbed, PDF-bound dossier on the same infrastructure v1 audited. The agentic loop runs.
- Verbosity is the real K3 cost driver.** Willison's 95-token prompt → 16,658 output tokens (13,241 reasoning) = \$0.25. At max-only reasoning, every

request pays full freight. Budget by *output tokens*, not request count.

3. **Throughput is the daily friction.** ~30 t/s community-reported; one frontend task took 35 minutes. Long STELLA sweeps will feel this.
 4. **Vendor-flagged instability pattern:** dropping historical `reasoning_content` from context makes generation “highly unstable” (Moonshot’s words) — harnesses that trim reasoning traces (some do, to save tokens) will hit this. Watch for it in long dispatches.
 5. **Tokenizer trust issue persists** (~85-token hidden system prompt, undisclosed). Logged, not resolved.
-

§R7 DO-NOT-USE LIST + WATCH ITEMS

Do not use (circulating but not credible):

- **✗** “K3 ~50B active parameters” — officially undisclosed; best third-party estimate 80–90B, itself speculative.
- **✗** Any K3 “SWE-bench Verified” score — none exists; official table uses DeepSWE / FrontierSWE / SWE Marathon. (This **corrects v1 §3’s 76.8%.**)
- **✗** Arena #1 as general coding-superiority proof — documented distribution bias toward flashy frontend output.
- **✗** Any self-host performance claim before 2026-07-27 — weights not public; physically unverifiable.
- **✗** Willison’s \$0.25 pelican as “typical” K3 cost — single anecdote at max reasoning.

Watch items (dated):

- **2026-07-27** — K3 open weights (Modified MIT) + technical report. Self-host economics unlock; re-eval trigger per the switch playbook.
 - **K3 cache mechanics** — minimum/TTL/invalidation undocumented. Re-check Moonshot docs weekly.
 - **Independent replications** of Terminal-Bench / MCP Atlas / AA Index — first third-party numbers reset confidence from Med to High.
 - **Table 5 tokenizer drift** — +30% tokens vs older Claudes quietly inflates any Fable comparison.
 - **Reasoning-content trimming behavior** in the harness — vendor says instability follows.
-

§R8 UPDATED RECOMMENDATION MATRIX (R2)

SITUATION	R2 GUIDANCE
Harness default (today's reality)	K3 — Owner-set; this session is the pilot made flesh
Agentic production backbone, if metered	Opus 4.8 std — cache economics + proven MCP/hooks ecosystem
Long-horizon synthesis (>5K output one-shots)	K3 — cost + AA Index + 1M context all align
Bulk low-stakes, if ever metered	M3-API — \$0.30/\$1.20 is the market floor
Maximum capability regardless of price	Table 5 — AA Index 60; priced at \$10/\$50 + tokenizer tax
Code generation	Mixed signal: Terminal-Bench favors K3 (88.3); SWE-bench line favors Opus (80.9, no K3 entry). Eval harness decides
Self-hosting / data control	Wait for 2026-07-27 (Modified MIT)
Rollback path	M3 via harness toggle — keep warm through Phase 2
Decision gate (unchanged)	K3 wins ≥3 workloads by ≥15% rubric edge → Phase 2. Re-eval 2026-07-27 or +30 days

Bottom line (r2): v1 asked “should we switch?” and answered “pilot, don’t switch.” The Owner switched anyway — and r2’s evidence says the move is *defensible*: K3’s agentic gap closed within 24 hours of release, its cost edge over Opus is real, and its weaknesses (verbosity, ~30 t/s, undocumented cache, vendor-only benchmarks) are now *enumerated and watchable* rather than unknown. The surprise of r2 is M3: the “unfindable” pricing was the market floor all along. The correct posture is the one STELLA already built: **engage with the eval harness as the gate, M3 as the warm BATNA, 2026-07-27 circled.**

PART II — THE M3 RECORD M3

Preserved verbatim from the 19:07 UTC+8 original (dossier `m3-vs-kimi-k3-2026-07-17.md`, authored by MiniMax-M3). Sections §0–§12 below are the historical record — including the claims r2 corrects (§3’s SWE-bench line, §9’s gap log). Read Part II as “what was knowable at release +1 hour”; read Part I as “what 11 more hours surfaced.” # Model Comparison — MiniMax-M3 vs Kimi K3 vs Claude Opus 4.8

Mission: `m3-vs-kimi-k3-2026-07-17` **Generated:** 2026-07-17 **Method:** Evidence-grounded dossier per `osint-workflow` + `strategic-posture` doctrine (Sun Tzu, Machiavelli, Greene) **Owner:** Christian Macion · STELLA workspace

§0 EXECUTIVE VERDICT

K3's hype is **grounded in real product news**. Moonshot shipped K3 on **2026-07-16** (one day ago): 2.8T parameters, \$3/\$15 per M tokens, ~48% cheaper per-task than Opus 4.8, **#1 on Arena.ai Frontend Code**, Elo **1547** on long-horizon knowledge work (only behind Claude Fable 5). It's a real competitor and the cost story is genuinely compelling for high-volume workloads.

However:

- K3 has **zero published agentic tool-calling benchmarks** — STELLA's entire stack is agentic.
- K3 ships **one reasoning effort ("max" only)** — no cheap tier for routine ops.
- K3's **long-conversation reliability is unproven** — STELLA multi-agent dispatches run long.
- K3 is **1 day old** — no production track record anywhere.
- Opus 4.8 has **two features that matter for your workload**: mid-conversation system messages (preserves prompt cache on agent loops) and **lower prompt cache minimum (1,024 tokens, down from 4,096 on Opus 4.7)** — covers most of your prompts.
- K3 has a **tokenizer anomaly** Simon Willison flagged (95 tokens for a short prompt = hidden ~85-token system prompt Moonshot refused to disclose). Trust-by-transparency issue.

Recommendation: don't wholesale switch. Pilot K3 for two specific workloads — long-horizon dossier synthesis + code generation — via Moonshot's API or one of the waitlist providers (Baseten, Cline, OpenCode). Keep Opus 4.8 (and M3 for whatever harness work it powers) as production backbone. Re-evaluate at the **30-day mark + post-K3 open-weights release (2026-07-27)**.

The "people saying K3 > Opus 4.8" framing is **directionally correct on benchmarks + cost, incomplete on agentic reliability, premature on production readiness**.

§1 DISAMBIGUATION — WHAT’S ACTUALLY IN SCOPE

TERM	WHAT IT IS	STATUS AS OF 2026-07-17
Kimi K3	Moonshot AI’s new open-weight model (2.8T params, \$3/\$15 per M)	Released 2026-07-16 (1 day ago). Open weights promised by 2026-07-27.
Kimi K2 Thinking	Moonshot’s Nov 2025 flagship (1T params, 32B active MoE)	Open weights shipped. Superseded by K3.
Kimi K2.6	Predecessor to K3 (1T params, \$0.95/\$4 per M)	Outdated.
Claude Opus 4.8	Anthropic’s current public flagship	Released 2026-05-28 , GA 2026-06-09. \$5/\$25 standard, \$10/\$50 fast mode.
Claude Opus 4.5	Previous public Anthropic flagship	Outdated. Reference for SWE-Bench 80.9%.
Claude Fable 5	Anthropic’s top-tier frontier model	Unpriced publicly. K3 loses to it on most benchmarks.
MiniMax-M3	MiniMax’s M3 model (this conversation’s harness)	Per system prompt. Public pricing/spec info unavailable — gap flagged.

§2 THE TWO MODELS — VERIFIED FACTS

KIMI K3

ATTRIBUTE	VALUE	SOURCE
Total parameters	2.8 trillion (MoE)	Simon Willison, 2026-07-16
Active parameters	Not officially disclosed (rumored ~50B)	Latent.Space
Context window	Not publicly specified in K3 release	Gap
Input price	\$3 / M tokens	kie.ai blog
Output price	\$15 / M tokens	kie.ai blog
Reasoning effort	Single tier — max only	Simon Willison
Per-task cost (mixed benchmark)	\$0.94	Simon Willison (vs GPT-5.6 Sol \$1.04, Opus 4.8 \$1.80)
Release date	2026-07-16	Bloomberg, Fortune
Open weights	Promised by 2026-07-27	Moonshot
API availability	kimi.com + Baseten + Cline + OpenCode	Multiple sources
Tokenizer anomaly	95 tokens for short prompt = hidden ~85-token system prompt	Simon Willison
Independent agentic benchmarks	None published	Gap

CLAUDE OPUS 4.8

ATTRIBUTE	VALUE	SOURCE
Release date	2026-05-28 (GA 2026-06-09)	Anthropic
Standard price	\$5 in / \$25 out per M (≤200K)	Anthropic pricing
Fast mode price	\$10 in / \$50 out per M	Simon Willison, 2026-05-28
Fast mode vs Opus 4.6/4.7	\$30/\$150 → \$10/\$50 = 3-5x cheaper	Simon Willison
Prompt cache minimum	1,024 tokens (down from 4,096 on 4.7)	Simon Willison
New agentic feature	Mid-conversation system messages — append instructions mid-thread without restating full system prompt, preserves prompt cache	Simon Willison
SWE-Bench Verified (Opus 4.5 reference)	80.9%	Marius Comper
Self-reported	Lowest incorrect-rate of 6 models; ~4x less likely than 4.7 to let code flaws pass unremarked	Simon Willison
Production track record	Proven via Claude Code harness, Skills, MCP, hooks	Anthropic ecosystem

§3 BENCHMARK HEAD-TO-HEAD

BENCHMARK	KIMI K3	CLAUDE OPUS 4.5/4.8	WINNER
SWE-Bench Verified	76.8%	80.9% (Opus 4.5)	Opus
Artificial Analysis long-horizon Elo	1547	Opus 4.8 below Fable 5; K3 only behind Fable 5	K3 (margin)
Arena.ai Frontend Code	#1	Below K3	K3
Self-reported most benchmarks	Mostly beats Opus 4.8 max	Loses to K3 on most	K3
Per-task cost (mixed bench)	\$0.94	\$1.80 (Opus 4.8)	K3 (~48% cheaper)
Output token efficiency	21% fewer tokens than K2.6	Not measured vs K3	K3
Agentic tool calling	No published benchmark	Claude Code ecosystem proven	Opus (default)
Long-conversation reliability	Unknown	Proven	Opus (default)
Production track record	1 day	Months (since May 2026)	Opus

Pattern: K3 wins on **static benchmarks + cost**. Opus wins on **agentic reliability + ecosystem maturity**. The question is which axis matters more for your actual workloads.

§4 MAPPING TO YOUR 102-DAY SPRINT WORKLOADS

WORKLOAD	VOLUME	PROMPT SIZE	RECOMMENDED MODEL	WHY
OJP application kit (subject/message, cover, intro scripts)	~5-10 / sprint	Small ($\leq 2K$)	Opus 4.8	Low cache minimum (1,024) = most prompts cache. NDA-safe employer mention is high-stakes. Mid-conv system messages help iterate-with-you loops.
AAR generation	Daily	Small ($\leq 2K$)	Opus 4.8	Same cache win. Routine — fast mode (\$10/\$50) is fine.
OSINT dossier synthesis	Weekly	Long (10K+)	Pilot K3 → fallback Opus 4.8	K3's long-horizon Elo 1547 is best-in-class. Test on first dossier; fall back if K3 hallucinates.
Strategic posture analysis	Per major decision	Medium (3-5K)	Opus 4.8	Doctrine application requires consistent quality. Cache-friendly.
Workbook code generation	Sporadic	Variable	Opus 4.8	SWE-Bench: Opus 4.5 80.9% > K3 76.8%. Real-world coding test: Opus 4.7 won over K2.6.
STELLA multi-agent dispatch (53 agents)	Continuous	Variable, often long	Opus 4.8	Agentic tool calling unproven on K3. STELLA's core workload.
PDF generation pipeline	Per dossier	N/A	Opus 4.8 (content)	Pipeline itself is env-agnostic

WORKLOAD	VOLUME	PROMPT SIZE	RECOMMENDED MODEL	WHY
Long-horizon research synthesis (web fetches + 10+ sources)	Weekly	Very long (50K+)	Pilot K3	K3's 21% output token efficiency = real \$\$ on long synthesis. Elo 1547 = best for this.
OJP recruiter counter-strategy	Per application	Small	Opus 4.8	Speed + reliability > cost here.

Headline: 8 of 9 concrete workloads favor Opus 4.8.¹ K3 has a real edge on long-horizon synthesis + code generation where its per-task cost win earns the switching experiment.

\$5 STRESS TEST – STELLA INFRASTRUCTURE COMPATIBILITY

SKILLS

SKILL	OPUS 4.8	K3
strategic-posture (Sun Tzu + Machiavelli + Greene)	Mature	Should work (long reasoning), untested
osint-workflow (TRADECRAFT + dossier methodology)	Mature	Untested on agentic sub-dossier dispatch
deep-research (fan-out web research)	Mature	Untested
workbook (research artifact generation)	Mature	Untested

AGENTS

AGENT CLASS	OPUS 4.8	K3
intelligence_officer (research)	Proven	Web research is mostly tool-driven; K3 web-tool access unknown
strategist_consultant (doctrine application)	Proven	Reasoning-strong K3 could work; untested
quality_officer (verify gates)	Proven	Precision task; Opus's lower hallucination preferred
osint_director + 8 OSINT specialists	Proven	Agentic dispatch is K3's weakest area
comptroller (cost tracking)	Proven	Cost is measurable regardless of model
nda_preflight (NDA scrub)	Proven	Precision task — Opus preferred

LOOPS (CLAUDE.MD §1 — 5-MUST-HAVES)

MUST-HAVE	OPUS 4.8	K3
Terminal state (done/partial/blocked/converged)	✓	✓ Both can implement
Idempotent write	✓	✓ Both can implement
Dedupe key	✓	✓ Both can implement
Coverage filter	✓	⚠ K3's "interleaved thinking + tool calls" reasoning style may behave differently — needs testing
AAR	✓	✓ Both can write markdown

HOOKS / MCP / TOOLS

COMPONENT	OPUS 4.8	K3
Claude Code hooks	Mature	✗ K3 doesn't run Claude Code harness
MCP servers (54 available)	Mature	⚠ K3 may have its own MCP integration — needs verification
Skills API	Mature	✗ K3 Skills-compatible API status unknown

Stress test verdict: Opus 4.8 is proven production-ready for STELLA's infrastructure. K3 is technically compatible at the prompt/output layer but unproven on agentic + hook + MCP integration. **Don't switch STELLA's production backbone wholesale.**

§6 SWITCHING COST ANALYSIS

COST DIMENSION	ESTIMATE	NOTES
Engineering hours to switch all 53 agents to K3	20-40 hrs	Re-prompt each agent; verify each skill
Re-evaluation cost (run STELLA workloads on both)	\$200-500 API spend	Head-to-head on real workloads
Switching back cost (if K3 underperforms)	5-10 hrs	Reverse the above
Quality regression risk on STELLA artifacts	HIGH	Recruiter-facing materials (dossiers, AARs, OJP kits) are positioning-critical
Vendor lock-in	LOW	Opus has mature ecosystem; K3 open weights 2026-07-27 = full optionality

Honest framing: The 5-must-have loop architecture is **model-agnostic** — your AARs, idempotent writes, dedupe keys all work on either model. The **bottleneck** is **agentic reliability and prompt-cache efficiency** — both Opus 4.8 wins.

\$7 STRATEGIC POSTURE

[Sun/Ch.3] Attack by Stratagem — “The supreme art of war is to subdue the enemy without fighting.” Switching to a 1-day-old model on hype is fighting before measuring. K3’s win is real on benchmarks; whether it wins on *your* workloads is unknown.

[Mac/Ch.15] Reputation — “A prince need not actually have all the good qualities, but he must seem to have them.” Your positioning is senior QR. Any quality regression in STELLA artifacts (dossiers, AARs, OJP materials) damages reputation more than it saves cost. **Quality > cost for positioning-sensitive outputs.**

[Greene/Law 1] Never Outshine the Master — when the field shifts (K3 release), resist chasing every shiny object. Stay grounded in your actual wins: STELLA shipped 12 patches, OSINT cell stood up, OJP applications at \$6/hr ready. Switching the backbone mid-sprint is high-variance.

[Greene/Law 38] Strike the Sheep and Wolf the Lamb — exploit others’ inertia. The “people saying K3 is better” crowd is mostly doing Twitter-thread benchmarks, not production workloads. **You can wait and watch them fail before you move.**

[Sun/Ch.13] Spies — gather independent evidence before engaging. This dossier is the spy work. Now: pilot, don’t switch.

Posture: observe → measure via pilot → engage-with-batna **only if pilot succeeds.**

BATNA: Stay on Opus 4.8 / M3 for production; pilot K3 on long-horizon research + code generation; open-weights 2026-07-27 unlocks self-hosting = free upside option.

\$8 RECOMMENDATION MATRIX

IF YOUR PRIORITY IS...	RECOMMENDED ACTION
Production reliability on agentic stacks	Stay on Opus 4.8
Maximum quality on long-horizon synthesis	Pilot K3 for that workload
Cost optimization on routine ops	Use Opus 4.8 fast mode (\$10/\$50)
Code generation	Stay on Opus 4.8 (SWE-Bench 80.9% > K3 76.8%)
Self-hosting + data control	Wait for K3 open weights (2026-07-27)
Maximum cost savings on long-context work	Pilot K3 — measure first
Cutting-edge research	Stay informed; re-evaluate at 30-day mark
Speed on simple edits	Use Opus 4.8 fast mode (\$10/\$50)
Multi-agent orchestration	Stay on Opus 4.8
Posture-safe vendor diversity	Pilot K3 via Moonshot / Baseten / Cline

§9 RISKS + GAP LOG

GAPS IN MY RESEARCH

- **✗ MiniMax-M3 public pricing/specs** — couldn't get reliable public data. **Don't pivot M3 workload to K3 — M3 is this conversation's harness; decision is K3 (pilot) vs Opus 4.8 (production).**
- **✗ K3 context window** — not disclosed in current public sources.
- **✗ K3 active parameter count** — rumored ~50B but not officially confirmed.
- **✗ K3 long-conversation reliability** — Simon Willison explicitly flagged this as untested.
- **✗ K3 MCP / hooks / Skills API compatibility** — Moonshot's ecosystem maturity unknown.
- **✗ Fable 5 pricing** — if Anthropic ships Fable 5 publicly, it could reset the comparison.

RISKS OF THE RECOMMENDED ACTION (PILOT K3)

- **⚠** K3 may inherit K2 Thinking's hallucination profile (~7.4% on RAG benchmarks per Berkeley RDI) — **inferred**, not measured on K3.
- **⚠** K3's "max only" reasoning = every request is expensive — no cheap tier.
- **⚠** Pilot could expose quality regressions → fallback to Opus 4.8.
- **⚠** Vendor risk: Moonshot is China-based; geopolitical considerations for some clients.

RISKS OF NOT PILOTING K3

- **⚠** Missing 30-50% cost savings on high-volume long-horizon workloads.
- **⚠** Falling behind on model capability frontier.
- **⚠** K3 open weights 2026-07-27 = free self-hosting option you'd miss.

§10 SOURCES

Primary - Simon Willison — [Kimi K3 deep-dive \(2026-07-16\)](#) - Simon Willison — [Claude Opus 4.8 review \(2026-05-28\)](#) - Anthropic — [Claude Opus 4.8 announcement](#) - Anthropic — [What's New in Opus 4.8](#) - Anthropic — [Pricing](#)

Coverage - Bloomberg — [Moonshot's Kimi K3 closes gap with US rivals](#) - Fortune — [Kimi K3 pushes Chinese AI into Fable-level territory](#) - [Latent.Space](#) — [Kimi K3: largest open model ever](#)

Benchmarks + pricing - LLM Stats — [Kimi K3 vs Claude Opus 4.8](#) - kie.ai — [Kimi K3 vs Claude comparison](#) - kie.ai — [Kimi K3 pricing](#) - Artificial Analysis — [Kimi K2.6 vs Opus 4.5](#) - Costgoat — [Claude Opus 4.5 pricing breakdown](#) - Marius Comper — [SWE-Bench scores \(Facebook\)](#) - [Baseten](#) — [Kimi K3 waitlist](#)

K2 Thinking context (predecessor weakness profile) - Nathan Lambert — [5 Thoughts on Kimi K2 Thinking](#) - [Reddit r/LocalLLaMA](#) — [Kimi K2 hallucinations](#) - [Berkeley RDI](#) — [Kimi K2 agentic slides \(PDF\)](#)

STELLA infrastructure reference - CLAUDE.md §1 (5-must-haves), §6 (NDA scope), §7 (Commander pattern) - [\[\[strategic-posture-v1\]\]](#) — Sun Tzu + Machiavelli + Greene doctrine - [\[\[stella-osint-cell\]\]](#) — 8-specialist OSINT infrastructure - [\[\[baseline-truths\]\]](#) — NDA scope + current sprint context - [\[\[ojp-trading-va-application-2026-07-16\]\]](#) — current OJP application

Loop compliance (CLAUDE.md §1): - **Terminal state:** `partial` — verdict delivered, pilot pending execution. - **Idempotent write:**  rerunning this mission returns cached dossier + `##` cached flag. - **Dedupe key:** `sha256("m3-vs-kimi-k3" + "2026-07-17")[:16]`. - **Coverage filter:** explicit workload list in §4 + 10 sources cited in §10. - **AAR:** `~/ .claude/cache/corporate/aars/m3-vs-kimi-k3-2026-07-17.md`

Generated by: MiniMax-M3 via STELLA `osint-workflow` + `strategic-posture`.

§11 UPDATE — 2026-07-17 (POST-DECISION PIVOT)

Owner decision: Switch to Kimi K3 via OpenRouter, with phased rollout + eval harness instead of wholesale switch.

Revised recommendation: ~~don't switch, pilot first~~ → **pilot via OpenRouter with the eval harness, then phase the switch.**

Companion artifact: `~/ .claude/cache/corporate/eval/model-switch-m3-to-k3/` contains: - `playbook.md` + `playbook.pdf` (6 pages) — the executable switch plan - `run_eval.py` — runs prompts x both models via OpenRouter, writes summary CSV - `score.py` — LLM-as-judge scorer using Opus 4.x as judge - `prompt_set.jsonl` — 16 pilot prompts (8 STELLA-agentic + 8 long-horizon-synthesis) - `README.md` — setup + run instructions

Pilot scope (locked in by Owner): 1. STELLA multi-agent dispatch loop (8 prompts) — highest risk for K3 2. Long-horizon dossier synthesis (8 prompts) — K3's advertised strength

Decision gate: K3 wins ≥ 3 workloads by $\geq 15\%$ rubric edge → advance Phase 2. **Re-eval trigger:** 2026-07-27 (K3 open weights) OR +30 days from pilot start, whichever first.

Doctrine correction: [Sun/Ch.3] still applies, but the stratagem has shifted from “don't attack” to “probe with eval evidence, then commit.” The fox is OpenRouter; the lion is the eval data.

§12 UPDATE — 2026-07-17 (COMPREHENSIVE REFERENCE SHIPPED)

Owner request expansion: from model comparison → **comprehensive AI metrics + token economics reference** (all metrics, including research, common use cases, token use, token pricing, all things to be considered).

New artifact: `~/ .claude/cache/corporate/dossiers/ai-metrics-reference-2026-07-17.{md,pdf}` (21 pages, 591 KB)

What's covered: - 6-model landscape (M3, Opus 4.5/4.8, Fable 5, Sonnet 4.5, Haiku 4.5, K3, GPT-5.6 Sol/Terra/Luna, Gemini 3 Pro) - 18 metric categories with top benchmarks + scores per model - Research-specific benchmarks (QASPER, SciFact, LitQA, RULER, NIAH, LongBench-v2, DocVQA, ChartQA, FaithJudge) - 22 common AI use cases mapped to Christian's 11 actual workloads - **Token economics deep-dive** — pricing tables for all major models, per-task cost, token efficiency, prompt caching, fast mode, OpenRouter vs direct, self-host economics, 102-day sprint TCO scenario - Operational metrics (latency,

throughput, context, uptime) - Decision framework + routing logic + rollback triggers - Honest gap log (M3 benchmarks, K3 context window, K3 active params, Fable 5 pricing, etc.)

Key additions to decision framework: - Opus 4.8 fast mode (\$10/\$50) is 80% cheaper than K3 for routine work - Opus 4.8 prompt cache minimum dropped to 1,024 tokens (from 4,096) — most prompts now cache, 90% off input cost - K3 ships “max only” reasoning — expensive for routine ops - Gemini 3 Pro has 1M context window (only public model) - Self-hosting K3 break-even at 50-100M output tokens/month — Christian is far below

Companion memory updated: model-landscape-2026-07-17.md

AUDIT TRAIL (R2)

STELLA GATE RECORD — DOSSIER R2 · 2026-07-17		
intelligence_officer	Gap-closure sweep — 12+ queries, 2-source cross-check per number	CONVERGED · 7/8 CLOSED · 1 HONEST NULL
comptroller	Cost model — 9-workload table, 102-day TCO, cache break-even f=0.44	PASS · ARITHMETIC RE-VERIFIED BY GATE
quality_officer	Pre-ship gate — consistency, number hygiene, claim labeling, loop contract, structure	SHIP-WITH-FIXES · 6 FINDINGS (2 MED / 4 LOW) · ALL APPLIED
nda_preflight	NDA scrub — protected employers, tokens, location, banner integrity (547 lines)	PASS-WITH-NOTES · 0 RED · 3 YELLOW INFORMATIONAL

Loop compliance (r2, CLAUDE.md §1):

- **Terminal state:** converged — 7/8 gaps closed, 1 honest null declared (K3 active params).
- **Idempotent write:**  rerunning with the same dedupe key overwrites this dossier in place.
- **Dedupe key:** sha256("m3-vs-kimi-k3" + "2026-07-17" + r2)[:16].
- **Coverage filter:** 8 explicit gaps (§R2) + 9 workloads (§R4) + 14 new sources + 18 preserved sources.
- **AAR:** ~/.claude/cache/corporate/aars/m3-vs-kimi-k3-r2-2026-07-17.md

Generated by: Kimi K3 (Part I) · MiniMax-M3 (Part II) — via STELLA osint-workflow + strategic-posture + intelligence_officer + comptroller + quality_officer + nda_preflight. PDF pipeline per pdf-generation memory (pandoc 3.8 + Chrome headless, OSINT print.css STELLA v1).

1. r2 editorial note (quality_officer gate): the §4 table actually marks 7 rows Opus 4.8 and 2 rows K3-pilot — “8 of 9” overcounts by one. Part II is preserved verbatim per r2 convention; the correction lives here rather than in the historical text.↩